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DISPLAY DEVICE AND METHOD OF MANUFACTURING THE SAME

Abstract

A display device and a manufacturing method thereof are discussed. The display device can include a substrate in which a plurality of sub pixels is defined, and a light emitting diode which is disposed in each of the plurality of sub pixels and has a first uneven structure, the light emitting diode includes a first semiconductor layer which has a first uneven structure formed on a bottom surface, an emission layer on the first semiconductor layer, a second semiconductor layer on the emission layer, a first electrode disposed on at least a part of a side surface of the first semiconductor layer, and a second electrode disposed on the second semiconductor layer. Accordingly, the first uneven structure is formed below the light emitting diode to improve the light extraction efficiency of the light emitting diode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Korean Patent Application No. 10-2024-0026207 filed on Feb. 23, 2024, in the Korean Intellectual Property Office, the entire contents of which is hereby expressly incorporated by reference into the present application.

BACKGROUND

Field

[0002] The present disclosure relates to a display device and a method of manufacturing the same, and more particularly to, a display device using a light emitting diode (LED) and a method of manufacturing the same.

Discussion of the Related Art

[0003] As display devices which are used for a monitor of a computer, a television, or a cellular phone, there are an organic light emitting display device (OLED) which is a self-emitting device and a liquid crystal display device (LCD) which requires a separate light source.

[0004] An applicable range of the display device is diversified to personal digital assistants as well as monitors of computers and televisions, and a display device with a large display area and a reduced volume and weight is being studied.

[0005] Further, in recent years, a display device including a light emitting diode (LED) is attracting attention as a next generation display device. Since the LED is formed of an inorganic material, rather than an organic reliability is excellent so that a lifespan thereof is longer than that of the liquid crystal display device or the organic light emitting display device. Further, the LED has a fast lighting speed, excellent luminous efficiency, and a strong impact resistance so that a stability is excellent and an image having a high luminance can be displayed.

SUMMARY OF THE DISCLOSURE

[0006] An object to be achieved by the present disclosure is to provide a display device which improves a yield of a self-assembly process of a light emitting diode and a method of manufacturing the display device.

[0007] Another object to be achieved by the present disclosure is to provide a display device in which a non-transferring defect which can be caused when a plurality of light emitting diodes is clustered together or bonded in an incorrect position is reduced, and a method of manufacturing the display device.

[0008] Still another object to be achieved by the present disclosure is to provide a display device which improves a light extraction efficiency by partially forming a ferromagnetic electrode having a low reflectance on a side surface of the light emitting diode, and a method of manufacturing the display device.

[0009] Still another object to be achieved by the present disclosure is to provide a display device which improves a light extraction efficiency by forming an uneven structure in a lower part and a side part of a light emitting diode, and a method of manufacturing the display device.

[0010] Still another object to be achieved by the present disclosure is to provide a display device in which a reflective partition is formed to enclose around a light emitting diode to reflect light extracted toward a side surface of the light emitting diode to an upper portion of the light emitting diode, and a method of manufacturing the display device.

[0011] Still another object to be achieved by the present disclosure is to provide a display device which reduces burrs of a first electrode and an encapsulation film of a light emitting diode, and a method of manufacturing the display device.

[0012] Still another object to be achieved by the present disclosure is to provide a display device which improves a light extraction efficiency of a light emitting diode to have a high efficiency and a high luminance, and a method of manufacturing the display device.

[0013] Objects of the present disclosure are not limited to the above-mentioned objects, and other objects, which are not mentioned above, can be clearly understood by those skilled in the art from the following descriptions.

[0014] According to an aspect of the present disclosure, a display device includes a substrate in which a plurality of sub pixels is defined; and a light emitting diode which is disposed in each of the plurality of sub pixels and has a first uneven structure, the light emitting diode includes: a first semiconductor layer which has a first uneven structure formed on a bottom surface; an emission layer on the first semiconductor layer; a second semiconductor layer on the emission layer; a first electrode disposed on at least a part of a side surface of the first semiconductor layer; and a second electrode disposed on the second semiconductor layer. Accordingly, the first uneven structure is formed below the light emitting diode to improve the light extraction efficiency of the light emitting diode.

[0015] According to an aspect of the present disclosure, a display device is provided. The display device comprises a substrate; and a plurality of sub pixels disposed on the substrate; wherein each of the plurality of sub pixels comprises a light emitting diode including: a first semiconductor layer having a bottom surface, a top surface and a side surface; an emission layer on the top surface; and a first electrode disposed on at least a part of the side surface; and the bottom surface has a first uneven structure. The first uneven structure can be an uneven structure formed by wet-etching using an alkaline etchant.

[0016] According to an aspect of the present disclosure, a method of manufacturing a display device includes placing a wafer and a plurality of light emitting diodes formed on one surface of the wafer in a chamber filled with a solution; separating the plurality of light emitting diodes from the wafer; placing an assembling substrate above the chamber; and moving the plurality of light emitting diodes to the assembling substrate by placing a magnet on the assembling substrate and self-assembling the plurality of light emitting diodes on the assembling substrate. Accordingly, in a state in which the plurality of light emitting diodes is disposed in the solution of the chamber, the light emitting diodes are separated from the wafer to relieve a shock to be applied to the plurality of light emitting diodes.

[0017] Other detailed matters of the example embodiments of the present disclosure are included in the detailed description and the drawings.

[0018] According to aspects of the present disclosure, a transferring process of a plurality of light emitting diodes is simplified to improve a yield of a display device.

[0019] According to aspects of the present disclosure, a defect that a light emitting diode is not transferred to a display panel caused when a plurality of light emitting diodes is clustered together or bonded in an incorrect position during a transferring process can be reduced.

[0020] According to aspects of the present disclosure, a ferromagnetic electrode having a low reflectance is formed on a side surface of a light emitting diode with a minimum area to improve a light extraction efficiency.

[0021] According to aspects of the present disclosure, an uneven structure is formed in a lower part and a side part of the light emitting diode to improve a light extraction efficiency.

[0022] According to aspects of the present disclosure, a light extraction efficiency of the light emitting diode is improved to implement a display device with a high efficiency and a high luminance.

[0023] According to aspects of the present disclosure, a display device which has a high efficiency

and a high luminance to be driven with a low power can be implemented.
[0024] According to aspects of the present disclosure, reflective partition which encloses around the light emitting diode is formed to reflect light extracted toward a side surface of the light emitting diode to an upper portion of the light emitting diode and a luminance of the display device can be improved.
[0025] According to aspects of the present disclosure, burrs of the first electrode and the encapsulation film of the light emitting diode can be reduced.
[0026] The effects according to the present disclosure are not limited to the contents exemplified above, and more various effects are included in the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0027] The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
[0028] FIG. **1** is a schematic diagram of a display device according to an example embodiment of the present disclosure;
[0029] FIG. **2**A is a partial cross-sectional view of a display device according to an example embodiment of the present disclosure;
[0030] FIG. **2**B is a perspective view of a tiling display device according to an example embodiment of the present disclosure;
[0031] FIG. **3** is a cross-sectional view of a sub pixel of a display device according to an example embodiment of the present disclosure;
[0032] FIGS. **4**A to **4**H are process diagrams for explaining a method of manufacturing a display device according to an example embodiment of the present disclosure;
[0033] FIG. **5** is a cross-sectional view of a display device according to another example embodiment of the present disclosure;
[0034] FIGS. **6**A to **6**C are schematic plan views of a light emitting diode of a display device according to various example embodiments of the present disclosure;
[0035] FIG. **7** is a cross-sectional view of a display device according to still another example embodiment of the present disclosure;
[0036] FIG. **8** is a cross-sectional view of a display device according to still another example embodiment of the present disclosure; and
[0037] FIG. **9** is a cross-sectional view of a display device according to still another example embodiment of the present disclosure.
DETAILED DESCRIPTION OF THE EMBODIMENTS
[0038] Advantages and characteristics of the present disclosure and a method of achieving the advantages and characteristics will be clear by referring to example embodiments described below in detail together with the accompanying drawings. However, the present disclosure is not limited to the example embodiments disclosed herein but will be implemented in various forms. The example embodiments are provided by way of example only so that those skilled in the art can fully understand the disclosures of the present disclosure and the scope of the present disclosure.
[0039] The shapes, sizes, ratios, angles, numbers, and the like illustrated in the accompanying drawings for describing the example embodiments of the present disclosure are merely examples, and the present disclosure is not limited thereto. Like reference numerals generally denote like elements throughout the disclosure. Further, in the following description of the present disclosure, a detailed explanation of known related technologies can be omitted to avoid unnecessarily obscuring the subject matter of the present disclosure. The terms such as “including,” “having,” and “consist

of” used herein are generally intended to allow other components to be added unless the terms are used with the term “only”. Any references to singular can include plural unless expressly stated otherwise.
[0040] Components are interpreted to include an ordinary error range even if not expressly stated.
[0041] When the position relation between two parts is described using the terms such as “on”, “above”, “below”, and “next”, one or more parts can be positioned between the two parts unless the terms are used with the term “immediately” or “directly”.
[0042] When an element or layer is disposed “on” another element or layer, another layer or another element can be interposed directly on the other element or therebetween.
[0043] Although the terms “first”, “second”, and the like are used for describing various components, these components are not confined by these terms. These terms are merely used for distinguishing one component from the other components, and may not define order or sequence. Therefore, a first component to be mentioned below can be a second component in a technical concept of the present disclosure.
[0044] Like reference numerals generally denote like elements throughout the disclosure.
[0045] A size and a thickness of each component illustrated in the drawing are illustrated for convenience of description, and the present disclosure is not limited to the size and the thickness of the component illustrated. Further, the term “can” fully encompasses all the meanings and coverages of the term “may.”
[0046] The features of various embodiments of the present disclosure can be partially or entirely adhered to or combined with each other and can be interlocked and operated in technically various ways, and the embodiments can be carried out independently of or in association with each other.
[0047] Hereinafter, various example embodiments of the present disclosure will be described in detail with reference to accompanying drawings. All the components of each display device according to all embodiments of the present disclosure are operatively coupled and configured.
[0048] FIG. **1** is a schematic diagram of a display device according to an example embodiment of the present disclosure. In FIG. **1**, for the convenience of description, among various components of a display device **100**, a display panel PN, a gate driver GD, a data driver DD, and a timing controller TC are illustrated.
[0049] Referring to FIG. **1**, the display device **100** includes the display panel PN including a plurality of sub pixels SP, the gate driver GD and the data driver DD which supply various signals to the display panel PN, and the timing controller TC which controls the gate driver GD and the data driver DD.
[0050] The gate driver GD supplies a plurality of scan signals to a plurality of scan lines SL according to a plurality of gate control signals supplied from the timing controller TC. Even though one gate driver GD is disposed to be spaced apart from one side of the display panel PN, the number of the gate drivers GD and the placement thereof are not limited thereto.
[0051] The data driver DD supplies a data voltage to a plurality of data lines DL according to a plurality of data control signals and image data supplied from the timing controller TC. The data driver DD converts the image data into a data voltage using a reference gamma voltage and can supply the converted data voltage to the plurality of data lines DL.
[0052] The timing controller TC aligns image data input from the outside to supply the image data to the data driver DD. The timing controller TC can generate a gate control signal and a data control signal using synchronization signals input from the outside, such as a dot clock signal, a data enable signal, and horizontal/vertical synchronization signals. Further, the timing controller TC supplies the generated gate control signal and data control signal to the gate driver GD and the data driver DD, respectively, to control the gate driver GD and the data driver DD.
[0053] The display panel PN is a configuration which displays images to the user and includes the plurality of sub pixels SP. In the display panel PN, the plurality of scan lines SL and the plurality of data lines DL intersect each other and the plurality of sub pixels SP is connected to intersections of

the scan lines SL and the data lines DL.

[0054] In the display panel PN, an active area AA (or display area) and a non-active area NA (or non-display area) can be defined. The non-active area NA can surround the active area AA entirely or only in part(s).

[0055] The active area AA is an area in which images are displayed in the display device **100**. In the active area AA, a plurality of sub pixels SP which configures a plurality of pixels PX and a pixel (PX) circuit for driving the plurality of sub pixels SP can be disposed. The plurality of sub pixels SP is a minimum unit which configures the active area AA and n sub pixels SP can form one pixel PX. In each of the plurality of sub pixels SP, a thin film transistor, etc. for driving the plurality of light emitting diodes **120** can be disposed. The plurality of light emitting diodes **120** can be defined in different ways depending on the type of the display panel PN. For example, when the display panel PN is an inorganic light emitting display panel PN, the light emitting diode **120** can be a light emitting diode (LED) or a micro light emitting diode (LED).

[0056] In the active area AA, a plurality of signal lines which transmits various signals to the plurality of sub pixels SP is disposed. For example, the plurality of signal lines can include a plurality of data lines DL which supplies a data voltage to each of the plurality of sub pixels SP and a plurality of scan lines SL which supplies a scan signal to each of the plurality of sub pixels SP. The plurality of scan lines SL extends to one direction in the active area AA to be connected to the plurality of sub pixels SP. The plurality of data lines DL extends to a direction different from the one direction in the active area AA to be connected to the plurality of sub pixels SP. In addition, in the active area AA, a low potential power line, a high potential power line, and the like, can be further disposed, but are not limited thereto.

[0057] The non-active area NA is an area where images are not displayed so that the non-active area NA can be defined as an area extending from the active area AA. In the non-active area NA, a link line which transmits a signal to the sub pixel SP of the active area AA, a pad electrode, or a driving IC, such as a gate driver IC or a data driver IC, can be disposed.

[0058] In the meantime, the non-active area NA can be located on a rear surface of the display panel PN, for example, a surface on which the sub pixels SP are not disposed or can be omitted, and is not limited as illustrated in the drawing.

[0059] In the meantime, a driver, such as a gate driver GD, a data driver DD, and a timing controller TC, can be connected to the display panel PN in various ways. For example, the gate driver GD can be mounted in the non-active area NA in a gate in panel (GIP) manner or mounted between the plurality of sub pixels SP in the active area AA in a gate in active area (GIA) manner.

[0060] For example, the data driver DD and the timing controller TC are formed in separate flexible film and printed circuit board. The display panel PN is electrically connected to the data driver DD and the timing controller TC by bonding the flexible film and the printed circuit board to the pad electrode formed in the non-active area NA of the display panel PN.

[0061] As another example, when the gate driver GD is mounted in the active area AA in the GIA manner and a side line SRL which connects the signal line on the front surface of the display panel PN to the pad electrode on a rear surface of the display panel PN is formed to bond the flexible film and the printed circuit board onto a rear surface of the display panel PN, the non-active area NA on the front surface of the display panel PN can be minimized. Therefore, when the gate driver GD, the data driver DD, and the timing controller TC are connected to the display panel PN as described above, a zero bezel in which there is no bezel can be substantially implemented, which will be described in more detail with reference to FIGS. **2**A and **2**B.

[0062] FIG. **2**A is a partial cross-sectional view of a display device according to an example embodiment of the present disclosure. FIG. **2**B is a perspective view of a tiling display device according to an example embodiment of the present disclosure.

[0063] In the non-active area NA of the display panel PN, a plurality of pad electrodes for transmitting various signals to the plurality of sub pixels SP is disposed. For example, in a non-active area NA on the front surface of the display panel PN, a first pad electrode PAD**1** which transmits a signal to the plurality of sub pixels SP is disposed. In a non-active area NA on the rear surface of the display panel PN, a second pad electrode PAD**2** which is electrically connected to a driving component, such as a flexible film and the printed circuit board, is disposed.

[0064] In this case, various signal lines connected to the plurality of sub pixels SP, for example, a scan line SL, a data line DL, or the like, extends from the active area AA to the non-active area NA to be electrically connected to the first pad electrode PAD**1**.

[0065] Further, the side line SRL is disposed along a side surface of the display panel PN. The side line SRL can electrically connect the first pad electrode PAD**1** on the front surface of the display panel PN and the second pad electrode PAD**2** on the rear surface of the display panel PN. Therefore, a signal from a driving component on the rear surface of the display panel PN can be transmitted to the plurality of sub pixels SP through the second pad electrode PAD**2**, the side line SRL, and the first pad electrode PAD**1**. Accordingly, a signal transmitting path is formed from the front surface of the display panel PN to the side surface and the rear surface to minimize an area of the non-active area NA on the front surface of the display panel PN.

[0066] Further, referring to FIG. **2**B, a tiling display device TD having a large screen size can be implemented by connecting a plurality of display devices **100**. At this time, as illustrated in FIG. **2**A, when the tiling display device TD is implemented using a display device **100** with a minimized bezel, a seam area in which an image between the display devices **100** is not displayed is minimized so that a display quality can be improved.

[0067] For example, the plurality of sub pixels SP forms one pixel PX and a distance D**1** between an outermost pixel PX of one display device **100** and an outermost pixel PX of another display device **100** adjacent to one display device can be implemented to be equal to a distance D**1** between pixels PX in one display device **100**. Accordingly, the distance between pixels PX between the display devices **100** is constantly configured to minimize the seam area.

[0068] However, FIGS. **2**A and **2**B are illustrative so that the display device **100** according to the example embodiment of the present disclosure can be a general display device with a bezel, but is not limited thereto.

[0069] Hereinafter, a display panel PN of a display device **100** according to an example embodiment of the present disclosure will be described in more detail.

[0070] FIG. **3** is a cross-sectional view of a sub pixel of a display device according to an example embodiment of the present disclosure.

[0071] Referring to FIG. **3**, the display panel PN of the display device **100** according to the example embodiment of the present disclosure includes a substrate **110**, a buffer layer **111**, a gate insulating layer **112**, a first interlayer insulating layer **113**, a second interlayer insulating layer **114**, a first planarization layer **115**, a passivation layer **116**, an adhesive layer **117**, a second planarization layer **118**, a third planarization layer **119**, a bank BB, a light shielding layer LS, a driving transistor DT, an auxiliary electrode LE, a power line PL, a plurality of reflective electrodes RE**1** and RE**2**, a plurality of connection electrodes CE**1** and CE**2**, and a light emitting diode **120**.

[0072] The substrate **110** is a component for supporting various components included in the display device **100** and can be formed of an insulating material. For example, the substrate **110** can be formed of glass or resin. Further, the substrate **110** can be configured to include a polymer or plastics or can be formed of a material having flexibility.

[0073] A light shielding layer LS is disposed in each of the plurality of sub pixels SP on the substrate **110**. The light shielding layer LS blocks light incident onto an active layer ACT of the driving transistor DT from the bottom of the substrate **110**. Light which is incident onto the active layer ACT of the driving transistor DT is blocked by the light shielding layer LS to reduce a leakage current.

[0074] A buffer layer **111** is disposed on the substrate **110** and the light shielding layer LS. The buffer layer **111** can reduce permeation of moisture or impurities through the substrate **110**. The

buffer layer **111** can be configured by a single layer or a double layer of silicon oxide (SiOx) or silicon nitride (SiNx), but is not limited thereto. However, the buffer layer **111** can be omitted depending on a type of substrate **110** or a type of transistor, but is not limited thereto.
[0075] The driving transistor DT is disposed on the buffer layer **111**. The driving transistor DT includes an active layer ACT, a gate electrode GE, a source electrode SE, and a drain electrode DE.
[0076] The active layer ACT is disposed on the buffer layer **111**. The active layer ACT can be formed of a semiconductor material, such as an oxide semiconductor, amorphous silicon, or polysilicon, but is not limited thereto.
[0077] The gate insulating layer **112** is disposed on the active layer ACT. The gate insulating layer **112** is an insulating layer which insulates the active layer ACT from the gate electrode GE and can be configured by a single layer or a double layer of silicon oxide (SiOx) or silicon nitride (SiNx), but is not limited thereto.
[0078] The gate electrode GE is disposed on the gate insulating layer **112**. The gate electrode GE can be configured by a conductive material, such as copper (Cu), aluminum (Al), molybdenum (Mo), nickel (Ni), titanium (Ti), chrome (Cr), or an alloy thereof, but is not limited thereto.
[0079] The first interlayer insulating layer **113** and the second interlayer insulating layer **114** are disposed on the gate electrode GE. In the first interlayer insulating layer **113** and the second interlayer insulating layer **114**, a contact hole through which the source electrode SE and the drain electrode DE are connected to the active layer ACT is formed. The first interlayer insulating layer **113** and the second interlayer insulating layer **114** are insulating layers for protecting a component below the first interlayer insulating layer **113** and the second interlayer insulating layer **114** and can be configured by a single layer or a double layer of silicon oxide (SiOx) or silicon nitride (SiNx), but are not limited thereto.
[0080] The source electrode SE and the drain electrode DE which are electrically connected to the active layer ACT are disposed on the second interlayer insulating layer **114**. The source electrode SE and the drain electrode DE can be configured by a conductive material, such as copper (Cu), aluminum (Al), molybdenum (Mo), nickel (Ni), titanium (Ti), chrome (Cr), or an alloy thereof, but are not limited thereto.
[0081] In the meantime, in the present disclosure, it is described that the first interlayer insulating layer **113** and the second interlayer insulating layer **114**, for example, a plurality of insulating layers is disposed between the gate electrode GE and the source electrode SE and the drain electrode DE. However, only one insulating layer can be disposed between the gate electrode GE and the source electrode SE and the drain electrode DE, but are not limited thereto.
[0082] Further, a plurality of insulating layers, such as the first interlayer insulating layer **113** and the second interlayer insulating layer **114**, is disposed between the gate electrode GE and the source electrode SE and the drain electrode DE. At this time, an electrode can be further: formed between the first interlayer insulating layer **113** and the second interlayer insulating layer **114**. The additionally formed electrode can form a capacitor with the other configuration disposed below first interlayer insulating layer **113** or above the second interlayer insulating layer **114**.
[0083] The auxiliary electrode LE is disposed on the gate insulating layer **112**. The auxiliary electrode LE is an electrode which electrically connects the light shielding layer LS below the buffer layer **111** to any one of the source electrode SE and the drain electrode DE on the second interlayer insulating layer **114**. For example, the light shielding layer LS is electrically connected to any one of the source electrode SE or the drain electrode DE through the auxiliary electrode LE so as not to operate as a floating gate. Therefore, fluctuation of a threshold voltage of the driving transistor DT caused by the floated light shielding layer LS can be reduced. Even though in the drawing, the light shielding layer LS is connected to the drain electrode DE, the light shielding layer LS can also be connected to the source electrode SE, but is not limited thereto.
[0084] The power line PL is disposed on the second interlayer insulating layer **114**. The power line PL is electrically connected to the light emitting diode **120** together with the driving transistor DT

to allow the light emitting diode **120** to emit light. The power line PL can be configured by a conductive material such as copper (Cu), aluminum (Al), molybdenum (Mo), nickel (Ni), titanium (Ti), chrome (Cr), or an alloy thereof, but is not limited thereto.

[0085] The first planarization layer **115** is disposed on the driving transistor DT and the power line PL. The first planarization layer **115** can planarize an upper portion of the substrate **110** on which the driving transistor DT is disposed. The first planarization layer **115** can be configured by a single layer or a double layer, and for example, can be formed of photoresist or an acrylic-based organic material, but is not limited thereto.

[0086] A plurality of reflective electrodes RE**1** and RE**2** which is spaced apart from each other is disposed on the first planarization layer **115**. The plurality of reflective electrodes RE**1** and RE**2** electrically connects the light emitting diode **120** to the power line PL and the driving transistor DT and serves as a reflector which reflects light emitted from the light emitting diode **120** to the upper portion of the light emitting diode **120**. The plurality of reflective electrodes RE**1** and RE**2** is formed of a conductive material having excellent reflecting property to reflect light emitted from the light emitting diode **120** toward the upper portion of the light emitting diode **120**. For example, the plurality of reflective electrodes RE**1** and RE**2** can be formed of a metal material having an excellent reflective property, such as aluminum (Al), silver (Ag), copper (Cu), palladium (Pd), or an alloy thereof, but is not limited thereto.

[0087] The plurality of reflective electrodes RE**1** and RE**2** includes a first reflective electrode RE**1** and a second reflective electrode RE**2**. The first reflective electrode RE**1** can electrically connect the driving transistor DT and the light emitting diode **120**. The first reflective electrode RE**1** can be connected to the source electrode SE or the drain electrode DE of the driving transistor DT through a contact hole formed in the first planarization layer **115**. Further, the first reflective electrode RE**1** can be electrically connected to the first electrode **124** and the first semiconductor layer **121** of the light emitting diode **120** through a first connection electrode CE**1** to be described below.

[0088] The second reflective electrode RE**2** can electrically connect the power line PL and the light emitting diode **120**. The second reflective electrode RE**2** is connected to the power line PL through a contact hole formed in the first planarization layer **115** and is electrically connected to a second electrode **125** and a second semiconductor layer **123** of the light emitting diode **120** through a second connection electrode CE**2** to be described below.

[0089] The first reflective electrode RE**1** or the second reflective electrode RE**2** are formed to overlap the entire area of the light emitting diode **120** below the light emitting diode **120** to reflect light from the light emitting diode **120**. For example, the first reflective electrode RE**1** is disposed so as to overlap the overall light emitting diode **120** and light directed to the bottom of the light emitting diode **120** can be reflected to the top of the substrate **110** by the first reflective electrode RE**1**. However, instead of the first reflective electrode RE**1**, the second reflective electrode REZ can be disposed so as to overlap the overall light emitting diode **120**, but is not limited thereto.

[0090] The passivation layer **116** is disposed on the plurality of reflective electrodes RE**1** and RE**2**. In the passivation layer **116**, a contact hole through which the first connection electrode CE**1** is connected to the first reflective electrode RE**1** and a contact hole through which the second connection electrode CE**2** is connected to the second reflective electrode RE**2** can be formed. The passivation layer **116** is an insulating layer which protects components below the passivation layer **116** and can be configured by a single layer or a double layer of silicon oxide (SiOx) or silicon nitride (SiNx), but is not limited thereto.

[0091] The adhesive layer **117** is disposed on the plurality of reflective electrodes RE**1** and RE**2** and the passivation layer **116**. The adhesive layer **117** is coated on the front surface of the substrate **110** to fix the light emitting diode **120** disposed on the adhesive layer **117**. For example, the adhesive layer **117** can be selected from any one of adhesive polymer, epoxy resist, UV resin, polyimide, acrylate, urethane, and polydimethylsiloxane (PDMS), but is not limited thereto.

[0092] The plurality of light emitting diodes **120** is disposed in each of the plurality of sub pixels

SP on the adhesive layer **117**. The plurality of light emitting diodes **120** is an element which emits light by a current. The plurality of light emitting diodes **120** can include a light emitting diode **120** which emits red light, green light, and blue light and implement various colored light including white by a combination thereof. For example, the plurality of light emitting diodes **120** can be a light emitting diode (LED) or a micro LED, but is not limited thereto.

[0093] The light emitting diode **120** includes a first semiconductor layer **121**, an emission layer **122**, a second semiconductor layer **123**, a first electrode **124**, a second electrode **125**, and an encapsulation film **126**. The first electrode **124** can include a first electrode layer **124***a* and a second electrode layer **124***b*.

[0094] The first semiconductor layer **121** is disposed on the adhesive layer **117** and the second semiconductor layer **123** is disposed on the first semiconductor layer **121**. The first semiconductor layer **121** and the second semiconductor layer **123** can be layers formed by doping n-type and p-type impurities into a specific material. For example, the first semiconductor layer **121** and the second semiconductor layer **123** can be layers doped with n-type and p-type impurities into a material such as gallium nitride (GaN), indium aluminum phosphide (InAlP), or gallium arsenide (GaAs). The p-type impurity can be magnesium (Mg), zinc (Zn), beryllium (Be), etc., and the n-type impurity can be silicon (Si), germanium (Ge), tin (Sn), etc., but are not limited thereto.

[0095] At this time, the first semiconductor layer **121** can include a part which protrudes to an outside of the second semiconductor layer **123** and the emission layer **122**. The first semiconductor layer **121** is formed by a part which overlaps the second semiconductor layer **123** and the emission layer **122** and a remaining part which encloses the part but does not overlap the second semiconductor layer **123** and the emission layer **122**. Therefore, a part of a top surface of the first semiconductor layer **121** can be exposed from the second semiconductor layer **123** and the emission layer **122**.

[0096] Further, a first uneven structure **121***a* can be formed on a bottom surface of the first semiconductor layer **121**. A roughness of the bottom surface of the first semiconductor layer **121** can be increased by forming the first uneven structure **121***a* on the bottom surface of the first semiconductor layer **121** of the light emitting diode **120**. Therefore, when the light emitting diode **120** is self-assembled, a defect that the light emitting diode **120** is adhered in an irregular position or the light emitting diode **120** is not picked up can be reduced. Further, a light extraction efficiency of the light emitting diode **120** can be improved, which will be described in more detail below.

[0097] The emission layer **122** is disposed between the first semiconductor layer **121** and the second semiconductor layer **123**. The emission layer **122** is supplied with holes and electrons from the first semiconductor layer **121** and the second semiconductor layer **123** to emit light. The emission layer **122** can be formed by a single layer or a multi-quantum well (MQW) structure, and for example, can be formed of indium gallium nitride (InGaN) or gallium nitride (GaN), but is not limited thereto.

[0098] The first electrode **124** is disposed on a side surface of the first semiconductor layer **121**. The first electrode **124** can be disposed on all or a part of the side surface of the first semiconductor layer **121**. Further, the first electrode **124** can be disposed on a top surface of the first semiconductor layer **121** which is exposed from the second semiconductor layer **123** and the emission layer **122**. The first electrode **124** can be disposed to extend from the side surface of the first semiconductor layer **121** to the top surface of the first semiconductor layer **121**. Further, an end portion of the first electrode **124** can be disposed on the top surface of the first semiconductor layer **121**. The first electrode **124** includes a first electrode layer **124***a* and a second electrode layer **124***b*.

[0099] The first electrode layer **124***a* is an electrode for electrically connecting the light emitting diode **120** and the driving transistor DT. The first electrode layer **124***a* can be configured by an opaque conductive material, such as titanium (Ti), gold (Au), silver (Ag), copper (Cu) or an alloy thereof, a transparent conductive material, such as indium tin oxide (ITO) or indium zinc oxide (IZO), or a combination of the opaque conductive material and the transparent conductive material. However, it is not limited thereto.

[0100] The second electrode layer **124***b* is disposed on the first electrode layer **124***a*. The second electrode layer **124***b* can be disposed so as to enclose the side surface of the first semiconductor layer **121** and the first electrode layer **124***a*. The second electrode layer **124***b* can be disposed on the side surface of first semiconductor layer **121** so as to cover the first electrode layer **124***a*. Further, an end portion of the second electrode layer **124***b* can be disposed so as to cover the top surface of the first semiconductor layer **121** and the end portion of the first electrode layer **124***a* disposed on the top surface of the first semiconductor layer **121**.

[0101] The second electrode layer **124***b* is an electrode which induces self-assembly of the light emitting diode **120** by forming an attractive force with the magnet MG during the self-assembly of the light emitting diode **120**. The second electrode layer **124***b* can be formed of a ferromagnetic material and for example, can be formed of a ferromagnetic material, such as iron (Fe), cobalt (Co), or nickel (Ni). Further, the second electrode layer **124***b* further includes a layer formed of a ferromagnetic material and an opaque conductive material having a high reflectance to be formed with a double-layered structure. In this case, some of light directed to a side surface, among light emitted from the light emitting diode **120** is reflected from the first electrode layer **124***a* and the second electrode layer **124***b* to be directed to the top of the light emitting diode **120**.

[0102] In the meantime, the first electrode layer **124***a* and the second electrode layer **124***b* disposed on a side portion of the first semiconductor layer **121** can be disposed in an area between the top surface of the first semiconductor layer **121** and the side surface of the first semiconductor layer **121**, due to a margin of a process of etching the first electrode layer **124***a* and the second electrode layer **124***b*. For example, one end and the other end of the first electrode layer **124***a* are disposed on a top surface and a side surface of the first semiconductor layer **121**. Further, one end and the other end of the second electrode layer **124***b* can be also disposed on a top surface and a side surface of the first semiconductor layer **121**. Specifically, it is difficult to perform the etching to remain the first electrode layer **124***a* and the second electrode layer **124***b* only on the side surface of the first semiconductor layer **121** due to the process error and a characteristic of process equipment.

[0103] Therefore, the etching process is performed to remain a part of the first electrode layer **124***a* and the second electrode layer **124***b* disposed between the top surface of the first semiconductor layer **121** and a lower edge of the first semiconductor layer **121**. Therefore, the first electrode layer **124***a* and the second electrode layer **124***b* can be also formed on the side surface of the first semiconductor layer **121**. Accordingly, the etching process is performed on the top surface of the first semiconductor layer **121** to form the first electrode layer **124***a* and the second electrode layer **124***b* also on the side surface of the first semiconductor layer **121**.

[0104] The second electrode **125** is disposed on the top surface of the second semiconductor layer **123**. The second electrode **125** is an electrode which electrically connects a second connection electrode CE**2** to be described below and the second semiconductor layer **123**. For example, the second electrode **125** is formed of a transparent conductive material, such as indium tin oxide (ITO) or indium zinc oxide (IZO), but is not limited thereto.

[0105] An encapsulation film **126** which encloses at least a part of the first semiconductor layer **121**, the emission layer **122**, the second semiconductor layer **123**, and the second electrode **125** is disposed. The encapsulation film **126** is formed of an insulating material to protect the first semiconductor layer **121**, the emission layer **122**, and the second semiconductor layer **123**. The encapsulation film **126** can cover an upper portion of the first semiconductor layer **121** adjacent to the emission layer **122**, a side surface of the emission layer **122**, a side surface of the second semiconductor layer **123**, and a part of an edge of the second electrode **125**. A part of the second electrode **125** is exposed from the encapsulation film **126** to electrically connect the second connection electrode CE**2** and the second electrode **125**.

[0106] Further, the encapsulation film **126** is disposed so as to cover an upper portion of the first semiconductor layer **121** adjacent to the emission layer **122**, for example, the side surface of the first semiconductor layer **121** adjacent to the emission layer **122** and a part of the top surface of the first semiconductor layer **121** which protrudes to the outside of the emission layer **122**. In this case, the encapsulation film **126** is disposed between an end portion of the second electrode layer **124***b* and an end portion of the first electrode layer **124***a* and a part of the top surface of the first semiconductor layer **121** and between an end portion of the second electrode layer **124***b* and an end portion of the first electrode layer **124***a* and an upper side surface of the first semiconductor layer **121**. By doing this, the second electrode layer **124***b* and the first electrode layer **124***a* are not in contact with the emission layer **122** and the second semiconductor layer **123** and a short circuit defect can be suppressed. The encapsulation film **126** can be formed of any one of insulating materials, such as silicon oxide (SiOx) or silicon nitride (SiNx), but is not limited thereto.
[0107] Next, the first connection electrode CE**1** is disposed on the adhesive layer **117** and the light emitting diode **120**. The first connection electrode CE**1** is an electrode which is disposed in each of the plurality of sub pixels SP to electrically connect the light emitting diode **120** and the driving transistor DT.
[0108] The first connection electrode CE**1** can be electrically connected to the first reflective electrode RE**1** through a contact hole formed in the adhesive layer **117** and the passivation layer **116**. Accordingly, the first connection electrode CE**1** can be electrically connected to any one of the source electrode SE and the drain electrode DE of the driving transistor DT through the first reflective electrode RE**1**.
[0109] Further, the first connection electrode CE**1** can be electrically connected to the first electrode layer **124***a* and the second electrode layer **124***b* of each of the plurality of light emitting diodes **120**. The first connection electrode CE**1** can be disposed so as to enclose the side surface of the light emitting diode **120**. The first connection electrode CE**1** can be disposed so as to enclose the side surface of the first semiconductor layer **121**, a top surface of the first semiconductor layer **121** protruding to the outside of the second semiconductor layer **123**, the first electrode layer **124***a*, and the second electrode layer **124***b*. The first connection electrode CE**1** can be disposed so as to enclose the lower side surface of the light emitting diode **120**. The first connection electrode CE**1** is in contact with the second electrode layer **124***b* to be electrically connected to the first electrode layer **124***a* and the first semiconductor layer **121**.
[0110] In summary, the first connection electrode CE**1** can be electrically connected to the first reflective electrode RE**1** and the driving transistor DT. The first connection electrode CE**1** is disposed so as to cover a side portion of the first semiconductor layer **121** in which the first electrode layer **124***a* and the second electrode layer **124***b* are formed to be electrically connected to the first electrode layer **124***a* and the second electrode layer **124***b*. Therefore, the driving transistor DT and the first electrode layer **124***a* and the first semiconductor layer **121** of the light emitting diode **120** can be electrically connected by means of the first connection electrode CE**1** and the first reflective electrode RE**1**.
[0111] The second planarization layer **118** is disposed on the first connection electrode CE**1** and the light emitting diode **120**. The third planarization s disposed on the second planarization layer **118**. The second planarization layer **118** and the third planarization layer **119** planarize an upper portion of the substrate **110** on which the light emitting diode **120** is disposed and fix the light emitting diode **120** onto the substrate **110** together with the adhesive layer **117**. The second planarization layer **118** overlaps a part of side surfaces of the plurality of light emitting diodes **120** to fix and protect the plurality of light emitting diodes **120**.
[0112] A thickness of the second planarization layer **118** can be smaller than a thickness of the first semiconductor layer **121** of the light emitting diode **120**. For example, the top surface of the second planarization layer **118** can be disposed below the emission layer **122**. When the display device **100** is manufactured, the self-alignment can be performed by adjusting the thickness of the second

planarization layer **118** to electrically connect the first connection electrode CE**1** only to the first semiconductor layer **121**. For example, a process of aligning the first electrode layer **124***a* of the light emitting diode **120** and the first connection electrode CE**1** is omitted and the first electrode **124** and the first connection electrode CEL are connected to each other by a self-alignment method, which will be described in more detail below.

[0113] The top surface of the third planarization layer **119** can be disposed to be higher than at least the emission layer **122** of the light emitting diode **120**. Further, the top surface of the third planarization layer **119** can be disposed at a height which is equal to or lower than the top surface of the second semiconductor layer **123**. For example, a top surface of the third planarization layer **119** can be disposed between the top surface of the emission layer **122** and the top surface of the second semiconductor layer **123** or on the same planar surface as the top surface of the second semiconductor layer **123**. At this time, when the display device **100** is manufactured, the self-alignment can be performed so that the second connection electrode CE**2** can be electrically connected only to the second semiconductor layer **123** using the third planarization layer **119**. For example, a process of aligning the second electrode **125** of the light emitting diode **120** and the second connection electrode CE**2** is omitted and the second electrode **125** and the second connection electrode CE**2** can be connected to each other by a self-alignment method, which will be described in more detail below.

[0114] Next, the second connection electrode CE**2** is disposed on the third planarization layer **119**. The second connection electrode CE**2** is an electrode for electrically connecting the light emitting diode **120** and the power line PL. The second connection electrode CE**2** can be electrically connected to the power line PL through a contact hole formed in the third planarization layer **119**, the second planarization layer **118**, the adhesive layer **117**, and the passivation layer **116**. Further, the second connection electrodes CE**2** are in contact with the top surface of the second electrode **125** which is exposed from the second planarization layer **118** to be electrically connected to the second electrode **125** of the light emitting diode **120** and the second semiconductor layer **123**. Accordingly, the second electrode **125** of the light emitting diode **120** and the second semiconductor layer **123** and the power line PL can be electrically connected to each other by means of the second connection electrode CE**2**.

[0115] The bank BB is disposed on the third planarization layer **119** and the second connection electrode CE**2**. The bank BB can be disposed to be spaced apart from the light emitting diode **120** with a predetermined interval. The bank BB can cover a part of the second connection electrode CE**2**. The bank BB is disposed at the boundary between the sub pixels SP which are adjacent to each other to reduce the mixture of light emitted from the light emitting diode **120** of each of the plurality of sub pixels SP. The bank BB can include an opaque material and a black component to reduce color mixture between the plurality of sub pixels SP and for example, can be formed of black resin, but is not limited thereto.

[0116] In the meantime, a protection layer can be further disposed on the bank BB. The protection layer is a layer for protecting configurations below the protection layer. The protection layer can be configured by a single layer or a double layer, and for example, configured by benzocyclobutene, a light-transmitting epoxy, a photoresist, or an acrylic-based organic material, but is not limited thereto.

[0117] Hereinafter, a method of manufacturing the display device **100** according to an example embodiment of the present disclosure will be described.

[0118] FIGS. **4**A to **4**H are process diagrams for explaining a method of manufacturing a display device according to an example embodiment of the present disclosure. FIGS. **4**A to **4**E are process diagrams for explaining a process of self-assembling the light emitting diode **120** and transferring the light emitting diode onto the display panel PN, and FIGS. **4**F to **4**H are process diagrams for explaining a process after transferring the light emitting diode **120** onto the display panel PN.

[0119] Referring to FIG. **4**A, a plurality of light emitting diodes **120** is formed on a wafer WF and is separated from the wafer WF. The plurality of light emitting diodes **120** which is separated from the wafer WF is disposed in a first chamber CB**1** filled with a solution ECT to remove a residual film below the first semiconductor layer **121** and form a first uneven structure **121***a*.

[0120] First, an epitaxial layer is grown on the wafer WF and is patterned into a plurality of parts to form a first semiconductor layer **121**, an emission layer **122**, and a second semiconductor layer **123** of the plurality of light emitting diodes **120**. Next, the second electrode **125** is formed on the second semiconductor layer **123** and an encapsulation film **126** covers the second electrode **125**, the second semiconductor layer **123**, and the emission layer **122**, and an upper portion of the first semiconductor layer **121**. Next, the first electrode layer **124***a* and the second electrode layer **124***b* are formed on a side portion of the first semiconductor layer **121** exposed from the encapsulation film **126** to form a plurality of light emitting diodes **120**. Accordingly, end portions of the first electrode layer **124***a* and the second electrode layer **124***b* can cover a part of the encapsulation film **126** disposed on the top surface of the first semiconductor layer **121**.

[0121] Further, before separating the plurality of light emitting diodes **120** from the wafer WF, a protection film PAS which covers the light emitting diode **120** can be formed. The protection film PAS is disposed so as to cover the light emitting diode **120** to protect the plurality of light emitting diodes **120** during the processing process. For example, the protection film PAS is disposed so as to cover an upper portion and a side portion of the light emitting diode **120** and cover the first electrode layer **124***a*, the second electrode layer **124***b*, and the encapsulation film **126**. The protection film PAS is formed of an insulating material, and for example, formed of an insulating material, such as silicon oxide (SiOx), but is not limited thereto.

[0122] Next, the plurality of light emitting diodes **120** on which the protection film PAS is formed can be separated from the wafer WF. The wafer WF and the plurality of light emitting diodes **120** can be separated by various methods, and for example, can be separated using a laser lift-off (LLO) method. For example, the laser lift-off (LLO) method, the laser is irradiated onto a rear surface of the wafer WF in a state in which the plurality of light emitting diodes **120** of the wafer WF and the first chamber CB**1** are disposed to be opposite to each other to separate the wafer WF and the light emitting diode **120**.

[0123] Next, the plurality of light emitting diodes **120** which is separated from the wafer WF is injected in the first chamber CB**1** filled with the solution ECT to remove a residual film on the bottom surface of the first semiconductor layer **121** and can form the first uneven structure **121***a*. Specifically, the plurality of light emitting diodes **120** can be separated in a state in which one surface of the wafer WF on which the plurality of light emitting diodes **120** is formed is disposed to be opposite to the first chamber CB**1**. The plurality of light emitting diodes **120** separated from the wafer WF can be injected into the first chamber CB**1** filled with the solution ECT. For example, the wafer WF and the light emitting diode **120** can be separated in a state in which the light emitting diode **120** is located on the solution ECT. As another example, the wafer WF and the light emitting diode **120** can be separated in a state in which the light emitting diode **120** is located in the solution ECT. Further, in a state in which the light emitting diode **120** is located in the solution ECT, for example, in a state in which at least a part of the wafer WF is disposed in the first chamber CB**1**, for example, if the light emitting diode **120** is separated by the laser lift-off (LLO) method, a shock applied to the light emitting diode **120** when the light emitting diode **120** and the wafer WF are separated can be relieved. At this time, the solution ECT can include various types of solvents or solutions. For example, the solution ECT can include de-ionized water (DI water) or an alkaline etchant such as KOH solution.

[0124] Next, a part of the light emitting diode **120** exposed from the protection film PAS can be wet-etched by the solution ECT. In this case, the solution ECT can include the KOH etchant. For example, a bottom surface of the first semiconductor layer **121** exposed from the protection film PAS can be wet-etched by the solution ECT. During this process, a gallium (Ga) residual film of the bottom surface of the first semiconductor layer **121** is removed and the first uneven structure

121*a* can be formed. During the process of separating the light emitting diode **120** and the wafer WF, laser can be irradiated onto the bottom surface of the light emitting diode **120** which is in contact with the wafer WF, for example, the bottom surface of the first semiconductor layer **121**. The first semiconductor layer **121** can be formed of gallium nitride (GaN). When the laser is irradiated onto the bottom surface of the first semiconductor layer **121**, a part of the lower portion of the first semiconductor layer **121** can be decomposed into gallium and nitrogen. Nitrogen is converted into gas and gallium can remain in a lower portion of the first semiconductor layer as a residual film. Accordingly, the light emitting diode **120** is injected into the solution ECT to remove the gallium residual film of the bottom surface of the semiconductor layer.

[0125] Further, the gallium residual film of the bottom surface of the first semiconductor layer **121** exposed from the protection film PAS and the bottom surface of the first semiconductor layer **121** is irregularly etched, simultaneously, to form the first uneven structure **121***a*. For example, the solution ECT is configured to be an alkaline etchant such as KOH etchant to irregularly etch the bottom surface of the first semiconductor layer **121** and the process time is adjusted to adjust a roughness of the bottom surface of the first semiconductor layer **121**.

[0126] Referring to FIG. **4**B, when the formation of the first uneven structure **121***a* below the first semiconductor layer **121** is completed, the plurality of light emitting diodes **120** is washed and the plurality of light emitting diodes **120** is self-assembled on an assembling substrate **1000**.

[0127] First, the plurality of light emitting diodes **120** is injected into a cleaning solution to wash the plurality of light emitting diodes **120**. For example, the cleaning solution can be de-ionized water (DI water). Accordingly, foreign materials generated from the etching process or the LLO process can be removed by the cleaning process.

[0128] Further, the plurality of light emitting diodes **120** can be injected into the second chamber CB**2** filled with fluid WT. The fluid WT can include various types of solvents or solutions. For example, the fluid WT can be formed of the same material as the solution ECT. As another example, the fluid WT can be de-ionized water (DI water) and a second chamber CB**2** filled with fluid WT can be a top-open type.

[0129] Next, the assembling substrate **1000** can be located on the second chamber CB**2** filled with the light emitting diode **120**. The assembling substrate **1000** can be disposed such that an organic layer OL on which a plurality of openings OLH of the assembling substrate **1000** is formed and the second chamber CB**2** are opposite to each other.

[0130] Next, a magnet MG can be located on the assembling substrate **1000**. The light emitting diodes **120** sinking on the bottom of the second chamber CB**2** or floating can move toward the assembling substrate **1000** by a magnetic force of the magnet MG.

[0131] At this time, the light emitting diode **120** can include magnetic materials to move by the magnetic field. For example, the second electrode layer **124***b* of the light emitting diode **120** includes a ferromagnetic material, such as iron (Fe), cobalt (Co), or nickel (Ni) to move the light emitting diode **120** to the magnet MG.

[0132] Referring to FIG. **4**C, the light emitting diode **120** which moves to the assembling substrate **1000** by the magnet MG can be self-assembled to the assembling substrate **1000** by an electric field formed between a plurality of assembly electrodes AE.

[0133] The assembling substrate **1000** includes an assembly substrate SUB, a plurality of assembly electrodes AE, an assembly insulating layer IL, and an organic layer OL.

[0134] A plurality of assembly electrodes AE is disposed on the assembly substrate SUB. The plurality of assembly electrodes AE includes a plurality of first assembly electrodes AE**1** and a plurality of second assembly electrodes AE**2**. The plurality of first assembly electrodes AE**1** and the plurality of second assembly electrodes AE**2** can be disposed to be spaced apart from each other with a predetermined interval. The plurality of first assembly electrodes AE**1** and the plurality of second assembly electrodes AE**2** can be alternately disposed. The plurality of first assembly electrodes AE**1** and the plurality of second assembly electrodes AE**2** are applied with different

voltages so that an electric field can be formed between the plurality of first assembly electrodes AE**1** and the plurality of second assembly electrodes AE**2**. Further, the plurality of light emitting diodes **120** can be self-assembled in an area between the plurality of first assembly electrodes AE**1** and the plurality of second assembly electrodes AE**2** using the electric field formed between the plurality of first assembly electrodes AE**1** and the plurality of second assembly electrodes AE**2**.
[0135] The assembly insulating layer IL is disposed on the plurality of assembly electrodes AE and the assembly substrate SUB. The assembly insulating layer IL protects the plurality of assembly electrodes AE from the fluid WT to suppress the defect such as corrosion of the plurality of assembly electrodes AE.
[0136] The organic layer OL including a plurality of openings OLH is disposed on the plurality of assembly electrodes AE. The organic layer OL is an insulating layer including the plurality of openings OLH in which the plurality of light emitting diodes **120** is seated and can be formed as a single layer or a double layer. For example, the thickness of the organic layer OL can be adjusted by configuring the organic layer OL as a single layer or a double layer.
[0137] Each of the plurality of openings OLH which is formed by opening a part of the organic layer OL is an area in which the plurality of light emitting diodes **120** is self-assembled. The plurality of openings OLH can be disposed so as to overlap an area between the first assembly electrode AE**1** and the second assembly electrode AE**2**. Further, each of the plurality of openings OLH can be formed in a position of the display panel PN corresponding to each of the plurality of sub pixels SP, later. The plurality of openings OLH can be disposed to correspond to the plurality of sub pixels SP one to one and the light emitting diode **120** which is self-assembled in the plurality of openings OLH can be aligned with the same interval as the plurality of sub pixels SP.
[0138] Further, a voltage is applied to the plurality of assembly electrodes AE to self-assemble the plurality of light emitting diodes **120** in the opening OLH of the organic layer OL. For example, different AC voltages are applied to the first assembly electrode AE**1** and the second assembly electrode AE**2** to form an electric field. The light emitting diode **120** is dielectrically polarized by the electric field to have a polarity. Further, the dielectrically polarized light emitting diode **120** can move or can be fixed to a specific direction by dielectrophoresis (DEP), for example, an electric field. Accordingly, the plurality of light emitting diodes **120** can be temporarily self-assembled in the opening OLH of the assembling substrate **1000** using dielectrophoresis.
[0139] In the meantime, during a process of self-assembling the plurality of light emitting diodes **120** in the assembling substrate **1000**, adsorption of the plurality of light emitting diodes **120** in a place other than the opening OLH, for example, a remaining part of the assembling substrate **1000** excluding the opening OLH, the first chamber CB**1**, or the second chamber CB**2** is minimized by forming the first uneven structure **121***a* on the bottom surface of the first semiconductor layer **121**. Specifically, gallium nitride (GaN) which forms the first semiconductor layer **121** is formed with a grid structure to have a polarity and the gallium nitride having polarity can be more easily combined in the other place. However, when the first uneven structure **121***a* is formed on the bottom surface of the first semiconductor layer **121**, the polarity of the bottom surface of the first semiconductor layer **121** is degraded and the bottom surface of the first semiconductor layer **121** can have a non-polarity or semi-polarity. Therefore, the first uneven structure **121***a* is formed on the bottom surface of the first semiconductor layer **121** so as to allow the bottom surface of the first semiconductor layer **121** to have semi-polarity or non-polarity. Further, the light emitting diode **120** can be suppressed from being attached to a place other than an area in the plurality of openings OLH or suppressed from being clustered. Further, the transferring process of the plurality of light emitting diodes **120** can be simplified. Accordingly, the first uneven structure **121***a* is formed in the plurality of light emitting diodes **120** to more easily transfer the plurality of light emitting diodes **120** onto the display panel PN and the non-transferring defect of the plurality of light emitting diodes **120** can be reduced.
[0140] Referring to FIG. **4**D, the plurality of light emitting diodes **120** of the assembling substrate

1000 is transferred onto the donor **2000**. Specifically, the assembling substrate **1000** and the donor **2000** can be aligned such that one surface of the assembling substrate **1000** in which the plurality of light emitting diodes **120** is disposed and one surface of the donor **2000** are opposite to each other. Further, the assembling substrate **1000** and the donor **2000** are bonded to transfer the light emitting diode **120** onto the donor **2000**. The donor **2000** is formed of a material having adhesiveness and upper portions of the plurality of light emitting diodes **120** are bonded to the donor **2000** to transfer from the assembling substrate **1000** to the donor **2000**.

[0141] When the plurality of light emitting diodes **120** is transferred from the assembling substrate **1000** to the donor **2000**, the first semiconductor layers **121** of the plurality of light emitting diodes **120** are in contact with the assembling substrate **1000**. At this time, the bottom surface of the first semiconductor layer **121** has a first uneven structure **121***a* and has a semi-polarity or a non-polarity so that the first semiconductor layer **121** of the light emitting diode **120** can be more easily separated from the assembling substrate **1000**. Accordingly, the defect that the plurality of light emitting diodes **120** is not transferred from the assembling substrate **1000** to the donor **2000** can be reduced.

[0142] Next, referring to FIGS. **4**E and **4**F, the plurality of light emitting diodes **120** of the donor **2000** is transferred onto the display panel PN.

[0143] First, the display panel PN has completed the process of forming the adhesive layer **117** and the display panel PN and the donor **2000** are aligned so that the adhesive layer **117** and the plurality of light emitting diodes **120** of the donor **2000** are opposite to each other. After disposing the donor **2000** such that the plurality of light emitting diodes **120** of the donor **2000** is opposite to the adhesive layer **117** of the display panel PN, the donor **2000** and the display panel PN are bonded to transfer the light emitting diode **120** on the donor **2000** onto the adhesive layer **117**. The plurality of light emitting diodes **120** disposed on the donor **2000** is disposed so as to correspond to the plurality of sub pixels SP so that all the light emitting diodes **120** on the donor **2000** are transferred onto the display panel PN at one time without selectively transferring the light emitting diodes **120**. The plurality of light emitting diodes **120** which is transferred onto the display panel PN is attached onto the adhesive layer **117** to be temporarily fixed.

[0144] Next, referring to FIG. **4**F, after transferring the light emitting diode **120** onto the adhesive layer **117** of the display panel PN, the protection film PAS of the light emitting diode **120** is removed and a contact hole is formed in the encapsulation film **126**. Specifically, the protection film PAS which covers the encapsulation film **126** of the light emitting diode **120** and the second electrode layer **124***b* can be removed. Further, a contact hole through which the second electrode **125** is exposed can be formed in the encapsulation film **126** by etching a part of the encapsulation film **126**. The second connection electrode CE**2** and the second electrode **125** to be formed later can be electrically connected through the contact hole of the encapsulation film **126**.

[0145] Next, a metal layer ML is formed on the light emitting diode **120** and the adhesive layer **117** and a second planarization material layer **118***m* is formed on the metal layer ML. The metal layer ML can be formed to cover all the light emitting diodes **120**. The metal layer ML can cover all the second electrode **125**, the encapsulation film **126**, and the second electrode layer **124***b*. Further, the metal layer ML can be connected to the first reflective electrode RE**1** through a contact hole formed in the adhesive layer **117** and the passivation layer **116**.

[0146] The second planarization material layer **118***m* formed on the metal layer ML is a material layer which forms the second planarization layer **118** and is formed as the second planarization layer **118** by an ashing process. Specifically, the second planarization material layer **118***m* is formed and the ashing process of reducing a thickness of the entire second planarization material layer **118***m* can be performed. Accordingly, the second planarization layer **118** can be formed by the ashing process of reducing the thickness of the second planarization material layer **118***m*. A top surface of the second planarization layer **118** can be disposed at a height lower than the emission layer **122**. The top surface of the second planarization layer **118** can be disposed to be lower than the bottom surface of the emission layer **122**. The second planarization layer **118** can have a thickness smaller than that of the first semiconductor layer **121**.

[0147] Next, the metal layer ML exposed from the second planarization layer **118** is etched to form the first connection electrode CE**1**. A part of the metal layer ML which encloses the second electrode **125**, the second semiconductor layer **123**, and the emission layer **122** of the light emitting diode **120** and is exposed from the second planarization layer **118** can be etched. In the drawings of the example embodiments, it is illustrated that in a portion in which first electrodes **124**, **524**, and **724** are formed on the first semiconductor layers **121**, **521**, and **721**, the thickness of the metal layer ML is equal to the thickness of the second planarization layer **118**. However, in order to suppress the metal layer ML formed on the first electrodes **124**, **524**, and **724** from being removed during the etching process, the second planarization layer **118** remains above the metal layer ML by the ashing process in a portion in which first electrodes **124**, **524**, and **724** are formed on the first semiconductor layers **121**, **521**, and **721**. Accordingly, a remaining part of the metal layer ML which is enclosed by the second planarization layer **118** remains to serve as the first connection electrode CE**1**. The second planarization layer **118** is used as a mask to form the first connection electrode CE**1**. Further, an uppermost part of the first connection electrode CE**1** can be disposed on the same planar surface as the top surface of the first planarization layer **115**.

[0148] Accordingly, instead of aligning the first connection electrode CE**1** so as to correspond to the first electrode layer **124***a* in the top and the side portion of the first semiconductor layer **121** of the light emitting diode **120**, only the metal layer ML exposed from the second planarization layer **118** having a smaller thickness than that of the first semiconductor layer **121** of the light emitting diode **120** is simply removed to self-align the first connection electrode CE**1**.

[0149] Next, referring to FIG. **4**G, a third planarization layer **119** is formed on the second planarization layer **118**. Specifically, the third planarization material layer **119***m* which covers the light emitting diode **120** is formed and the ashing process is formed on the third planarization material layer **119***m* to form the third planarization layer **119**. An upper portion of the third planarization material layer **119***m* is entirely removed by the ashing process to expose the second electrode **125** of the light emitting diode **120** from the third planarization layer **119**.

[0150] Finally, referring to FIG. **4**H, the second connection electrode CE**2** is formed on the third planarization layer **119**. The second connection electrode CE**2** is in contact with the second electrode **125** of the light emitting diode **120** exposed from the third planarization layer **119**. Further, the second connection electrode CE**2** can be connected to the second reflective electrode RE**2** through the contact hole formed in the third planarization layer **119**, the second planarization layer **118**, the adhesive layer **117**, and the passivation layer **116**. Accordingly, the second electrode **125** of the light emitting diode **120** and the second reflective electrode RE**2** and the power line PL are electrically connected to each other by means of the second connection electrode CE**2**.

[0151] Next, the bank BB is formed on the second connection electrode CE**2** and the third planarization layer **119**. The bank BB is formed in a surrounding area of the light emitting diode **120** to reduce color mixture between the plurality of sub pixels SP.

[0152] In the meantime, the contact hole through which the second connection electrode CE**2** and the second reflective electrode RE**2** are in contact can be formed in various ways. For example, in each of the forming processes of the passivation layer **116**, the adhesive layer **117**, the second planarization layer **118**, and the third planarization layer **119**, the contact hole is partially formed and the contact holes of the layers are connected to finally form a contact hole through which the second connection electrode CE**2** and the second reflective electrode RE**2** are connected. Further, the contact holes are simultaneously formed in several layers and for example, the contact holes are partially formed in the passivation layer **116** and the adhesive layer **117**. When the second planarization layer **118** and the third planarization layer **119** are formed by the subsequent process, the contact hole is partially formed. Further, after forming the third planarization layer **119**, the contact holes can be formed entirely from the third planarization layer **119** to the passivation layer

116 at one time. A process and an order of forming the contact holes can be modified in various forms, but are not limited thereto.

[0153] In the display device **100** according to the example embodiment of the present disclosure, the first uneven structure **121***a* is formed on the bottom surface of the first semiconductor layer **121** to reduce the attachment of the plurality of light emitting diodes **120** in an incorrect position. Specifically, during the process of forming the first uneven structure **121***a* on the bottom surface of the light emitting diode **120** or self-assembling the light emitting diode **120**, a defect that the light emitting diode **120** is attached in a place other than the opening OLH of the assembling substrate **1000** can occur. Specifically, the first semiconductor layer **121** of the light emitting diode **120** is formed of gallium nitride having polarity to be easily bonded to the other place. Therefore, the first uneven structure **121***a* is formed on the bottom surface of the first semiconductor layer **121** of the light emitting diode **120** to allow the bottom surface of the first semiconductor layer **121** to have a non-polarity or a semi-polarity. Accordingly, the defect that the plurality of light emitting diodes **120** is adhered to the first chamber CB**1** and the second chamber CB**2** or adhered to a place other than the opening OLH of the assembling substrate **1000** can be suppressed. Further, the light emitting diode **120** seated in the plurality of openings OLH is easily separated from the assembling substrate **1000** to be transferred onto the donor **2000**. Accordingly, the first uneven structure **121***a* is formed below the first semiconductor layer **121** of the light emitting diode **120** to reduce the non-transferring defect of the light emitting diode **120** or the defect that the light emitting diode **120** is adhered to an incorrect position and improve the manufacturing yield of the display device **100**.

[0154] FIG. **5** is a cross-sectional view of a display device according to another example embodiment of the present disclosure. FIGS. **6**A to **6**C are schematic plan views of a light emitting diode of a display device according to various example of the present disclosure. The difference between a display device **500** of FIGS. **5** to **6**C and the display device **100** of FIGS. **1** to **3** is a light emitting diode **520**, but other configurations are substantially the same, so that a redundant description will be omitted or may be briefly provided.

[0155] Referring to FIG. **5**, a first semiconductor layer **521** of a light emitting diode **520** can include a part below an emission layer **522** and a part protruding from the second semiconductor layer in one direction. A part of a side surface of the first semiconductor layer **521** can be disposed on the same line as a side surface of the emission layer **522** and a side surface of the second semiconductor layer **523**. Further, the other part of the side surface of the first semiconductor layer **521** in a protruding part of the first semiconductor layer **521** protrudes outside more than a side surface of the emission layer **522** and a side surface of the second semiconductor layer **523** to be disposed on a different planar surface therefrom. The protruding part of the first semiconductor layer **521** can be a part formed for a process margin when the first electrode layer **524***a* and the second electrode layer **524***b* are formed.

[0156] The light emitting diode **520** includes a first electrode layer **524***a* which covers a side portion of the first semiconductor layer **521** and a second electrode layer **524***b* which covers the first electrode layer **524***a*. The first electrode layer **524***a* and the second electrode layer **524***b* can be disposed to cover only a part of the side surface of first semiconductor layer **521**. The first electrode layer **524***a* and the second electrode layer **524***b* can be disposed on a side surface of the protruding part of first semiconductor layer **521**, among the side surface of first semiconductor layer **521**. Further, the first electrode layer **524***a* and the second electrode layer **524***b* can be disposed on a top surface of the protruding part of first semiconductor layer **521**.

[0157] For example, one end portion of the second electrode layer **524***b* and one end portion of the first electrode layer **524***a* can be disposed on a top surface of the first semiconductor layer **521** which protrudes to the outside of the second semiconductor layer **523** and the emission layer **522**. Further, the other end portion of the second electrode layer **524***b* and the other end portion of the first electrode layer **524***a* can be disposed to be spaced apart from a lower edge of the side surface of the first semiconductor layer **521**. The other end portions of the first electrode layer **524***a* and the second electrode layer **524***b* can be disposed on a lower portion of first semiconductor layer **521** to be disposed to be spaced apart from the lower edge of the first semiconductor layer **521**. In a spaced part of the lower edge of the first semiconductor layer **521** and the first electrode **524**, the first connection electrode CE**1** can be in contact with the bottom surfaces of the first semiconductor layer **521** and the first electrode **524**.

[0158] The encapsulation film **526** is formed on a side surface of the first semiconductor layer **521** in which the first electrode layer **524***a* and the second electrode layer **524***b* are not formed to protect the first semiconductor layer **521**.

[0159] Further, an area of the emission layer **522** can expand to the side surface of the first semiconductor layer **521** in which the first electrode layer **524***a* and the second electrode layer **524***b* are not formed. For example, a part of the top surface of the first semiconductor layer **521** needs to be exposed to the emission layer **522** for the process margin of the first electrode layer **524***a* and the second electrode layer **524***b*. However, in a part in which the first electrode layer **524***a* and the second electrode layer **524***b* are not formed, the emission layer **522** can be disposed on the top surface of the first semiconductor layer **521**. Accordingly, the area of the emission layer **522** of the light emitting diode **520** can be increased and the luminous efficiency of the light emitting diode **520** can be improved.

[0160] Referring to FIGS. **6**A to **6**C, a placement area of the first electrode layer **524***a* and the second electrode layer **524***b* can be designed in various forms in consideration of the shape of the light emitting diode **520**. For example, referring to FIG. **6**A, when the overall planar surface shape of the light emitting diode **520**′ is an oval shape, the first electrode layer **524***a* and the second electrode layer **524***b* can be disposed so as to cover only a part of the side portion of the first semiconductor layer **521**. At this time, the planar surface shape of the first electrode layer **524***a* and the second electrode layer **524***b* can be an arc shape.

[0161] For example, referring to FIG. **6**B, when the overall planar surface shape of the light emitting diode **520**″ is a rectangular shape, the first electrode layer **524***a* and the second electrode layer **524***b* can be disposed so as to correspond to one side of the first semiconductor layer **521**.

[0162] For example, referring to FIG. **6**C, when the overall planar surface shape of the light emitting diode **520**‴ is a rectangular shape and the emission layer **522**, the second semiconductor layer **523**, and the second electrode have a planar surface shape which has a groove formed in at least a part, the first electrode layer **524***a* and the second electrode layer **524***b* are disposed so as to correspond to one side of the first semiconductor layer **521** and can be also disposed in the grooves of the emission layer **522**, the second semiconductor layer **523**, and the second electrode **525**.

[0163] However, the planar surface shape of the light emitting diode **520** and the placement area of the first electrode layer **524***a* and the second electrode layer **524***b* can be designed in various forms, but are not limited thereto.

[0164] Accordingly, in the display device **500** according to another example embodiment of the present disclosure, the first electrode layer **524***a* and the second electrode layer **524***b* are formed only in a part of the side surface of the first semiconductor layer **521** to improve the light extraction efficiency. Light emitted from the emission layer **522** travels in various directions so that there can be light which travels toward the first semiconductor layer **521**. At this time, some light which is directed toward the side surface of the first semiconductor layer **521** can travel to the outside of the light emitting diode **520** through a part in which the first electrode layer **524***a* and the second electrode layer **524***b* are not disposed. Specifically, light may not transmit the second electrode layer **524***b* which is formed of an opaque ferromagnetic material for the sake of self-assembly of the light emitting diode **520**. Further, the second electrode layer **524***b* is formed of a ferromagnetic material having a low reflectance so that the larger the area of the second electrode layer **524***b*, the lower the light extraction efficiency of the light emitting diode **520**. Therefore, the second electrode layer **524***b* is disposed on a part of the side surface of the first semiconductor layer **521** with a minimum area so that more light can be extracted to the side surface of the first semiconductor layer **521**. Accordingly, the first electrode layer **524***a* and the second electrode layer **524***b* are formed only in a part of the side surface of the light emitting diode **520** to improve a light output efficiency to the side surface of the first semiconductor layer **521**. Further, the display device **500** which has a high efficiency and a high luminance to be driven at a low power can be implemented.

[0165] In the display device **500** according to another example embodiment of the present disclosure, the end portions of the first electrode layer **524***a* and the second electrode layer **524***b* are disposed to be higher than a lower edge of the first semiconductor layer **521** to suppress burrs from the end portions of the first electrode layer **524***a* and the second electrode layer **524***b*. During a process of forming the first uneven structure **521***a* on the bottom surface of the first semiconductor layer **521** using a solution ECT including etchant or fluid WT, burrs can be generated between the lower edge of the first semiconductor layer **521** and the end portion of the first electrode layer **524***a*. Further, lifting can occur between the first electrode layer **524***a* and the first semiconductor layer **521**. Accordingly, the end portions of the first electrode layer **524***a* and the second electrode layer **524***b* are disposed to be higher than the bottom surface of the first semiconductor layer **521** to suppress burrs in the first electrode layer **524***a* and the second electrode layer **524***b* and reduce the defect of the light emitting diode **520**.

[0166] FIG. **7** is a cross-sectional view of a display device according to still another example embodiment of the present disclosure. The difference between a display device **700** of FIG. **7** and the display device **500** of FIGS. **5** and **6** is a light emitting diode **720**, but other configurations are substantially the same, so that a redundant description will be omitted or may be briefly provided.

[0167] Referring to FIG. **7**, a first semiconductor layer **721** of the light emitting diode **720** includes a first uneven structure **721***a* of a bottom surface of the first semiconductor layer **721** and a second uneven structure **721***b* of a bottom surface of the first semiconductor layer **721**. The first uneven structure **721***a* can be formed on the entire bottom surface of the first semiconductor layer **721**. The second uneven structure **721***b* can be formed on a part of the side surface of the first semiconductor layer **721** on which the first electrode layer **724***a* and the second electrode layer **724***b* are not disposed.

[0168] Further, the part in which the second uneven structure **721***b* is formed can be exposed from the encapsulation film **726**. The side surface of the first semiconductor layer **721** can be formed by a side surface part in which the first electrode layer **724***a* and the second electrode layer **724***b* are disposed, a side surface part which is exposed from the encapsulation film **726** and has a second uneven structure **721***b*, and a side surface part which is covered by the encapsulation film **726**.

[0169] In the meantime, both the encapsulation film **726** and the protection film PAS may not be formed in a part in which the second uneven structure **721***b* is formed during the manufacturing process of the light emitting diode **720**. After forming the first semiconductor layer **721**, the emission layer **722**, the second semiconductor layer **723**, the first electrode layer **724***a*, the second electrode layer **724***b*, and the second electrode **725** of the light emitting diode **720** on the wafer WF, the encapsulation film **726** and the protection film PAS which cover them can be formed. At this time, the encapsulation film **726** and the protection film PAS are omitted on a part on which the second uneven structure **721***b* will be formed and the side surface of the first semiconductor layer **721** exposed from the encapsulation film **726** and the protection film PAS is exposed to the solution ECT including etchant to form the second uneven structure **721***b*.

[0170] Accordingly, in the display device **700** according to still another example embodiment of the present disclosure, the end portion of the encapsulation film **726** is disposed to be higher than the lower edge of the first semiconductor layer **721** to suppress burrs from the end portion of the encapsulation film **726**. For example, the end portion of the encapsulation film **726** is disposed to be higher than the bottom surface of the first semiconductor layer **721** to be spaced apart from the bottom surface of the first semiconductor layer **721**. The encapsulation film **726** may not be formed in a lower part of the first semiconductor layer **721**. When the first uneven structure **721***a* is formed on the bottom surface of the first semiconductor layer **721**, burrs can be generated between the first uneven structure **721***a* of the bottom surface of the first semiconductor layer **721** and the end portion of the encapsulation film **726** and thus lifting can be generated in the encapsulation film **726**. Therefore, the end portion of the encapsulation film **726** is formed to be higher than the bottom surface of the first semiconductor layer **721** to suppress burrs in the end portion of the encapsulation film **726** and reduce the lifting of the encapsulation film **726** to improve the reliability of the light emitting diode **720**.

[0171] In the display device **700** according to still another example embodiment of the present disclosure, the second uneven structure **721***b* is formed on the side surface of the first semiconductor layer **721** exposed from the encapsulation film **726** to improve the light extraction efficiency to the side portion of the light emitting diode **720**. Light emitted from the emission layer **722** travels to various directions and some light can travel to the side surface or the bottom surface of the first semiconductor layer **721** in which the first electrode layer **724***a* and the second electrode layer **724***b* are not formed. At this time, light which travels to the side surface of the first semiconductor layer **721** in which the first electrode layer **724***a* and the second electrode layer **724***b* are not formed can be more easily extracted to the outside of the first semiconductor layer **721** by the second uneven structure **721***b*. Further, light which travels to the bottom surface of the first semiconductor layer **721** can be more easily extracted to the outside of the first semiconductor layer **721** by the first uneven structure **721***a*. Accordingly, some light which travels from the emission layer **722** to the first semiconductor layer **721** is scattered and reflected by the first uneven structure **721***a* and the second uneven structure **721***b* to be more easily extracted to the outside of the light emitting diode **720**. Therefore, the light extraction efficiency of the light emitting diode **720** and the luminance of the display device **700** can be improved.

[0172] FIG. **8** is a cross-sectional view of a display device according to still another example embodiment of the present disclosure. The difference between a display device **800** of FIG. **8** and the display device **700** of FIG. **7** is that a reflective partition **830** is further included, but the other configuration is substantially the same, so that a redundant description will be omitted or may be briefly provided.

[0173] Referring to FIG. **8**, a reflective partition **830** which encloses the light emitting diode **720** is disposed in each of the plurality of sub pixels SP. The reflective partition **830** can be disposed around the light emitting diode **720** to enclose the light emitting diode **720**. For example, in the planar surface view, the reflective partition **830** can be formed with a closed-loop ring shape which encloses the light emitting diode **720**. The reflective partition **830** can be disposed on the adhesive layer **117** and the first connection electrode CE**1**.

[0174] The reflective partition **830** includes a partition layer **831** and a reflective layer **832**. The partition layer **831** can be disposed on the adhesive layer **117** and the first connection electrode CE**1** so as to enclose the plurality of light emitting diodes **720**. A height of the reflective layer **832** which is formed of a metal material can be adjusted by adjusting a thickness of the partition layer **831**. The partition layer **831** can be formed of an insulating material, and for example, can be formed of photoresist or an acrylic-based organic material, but is not limited thereto. Further, the partition layer **831** is configured by a single layer or a double layer to adjust a thickness of the partition layer **831**.

[0175] The reflective layer **832** can cover a top surface and a side surface of the partition layer **831**. The reflective layer **832** is formed on a surface of the partition layer **831** to reflect light which is directed to the side surface of the light emitting diode **720**, among light emitted from the light emitting diode **720**, to the upper direction of the light emitting diode **720**. The reflective layer **832** is disposed so as to enclose the lower portion and the side portion of the light emitting diode **720** together with the first reflective electrode RE**1** and the second reflective electrode RE**2** to reflect the light emitted from the light emitting diode **720** to the top of the light emitting diode **720**. The reflective layer **832** can be formed of a metal material having an excellent reflective property, such as aluminum (Al), silver (Ag), copper (Cu), palladium (Pd), or an alloy thereof, but is not limited thereto.

[0176] In the meantime, a part of the reflective layer **832** is in contact with the first connection electrode CE**1** to be electrically connected. At this time, the reflective layer **832** is enclosed by the second planarization layer **118** and the third planarization layer **119** to be separated from the other configuration, such as a second connection electrode CE**2**, so that the short defect due to the reflective layer **832** can be suppressed. Specifically, the third planarization layer **119** covers a top of the reflective layer **832** to insulate the reflective layer **832** and the second connection electrode CE**2** from each other.

[0177] During the manufacturing process of the display device **800**, a metal layer ML for forming the first connection electrode CE**1** is formed and the partition layer **831** and the reflective layer **832** can be sequentially formed on the metal layer ML and the adhesive layer **117**. Further, after forming the second planarization layer **118** on the reflective partition **830**, a part of the metal layer ML exposed from the second planarization layer **118** is etched to form the first connection electrode CE**1**. Finally, the third planarization layer **119**, the second connection electrode CE**2**, and the bank BB are sequentially formed on the reflective partition **830** and the second planarization layer **118** to complete the manufacturing process of the display device **800**.

[0178] Accordingly, in the display device **800** according to still another example embodiment of the present disclosure, a reflective partition **830** which encloses the light emitting diode **720** is formed to improve the light extraction efficiency of the light emitting diode **720**. Some of light emitted from the light emitting diode **720** travels to the bottom surface of the first semiconductor layer **721** or the side surface of the first semiconductor layer **721** in which the first electrode layer **724***a* and the second electrode layer **724***b* are not formed. Light which travels to the side surface of the first semiconductor layer **721** is scattered and reflected by the first uneven structure **721***a* and the second uneven structure **721***b* to be extracted to the outside of the light emitting diode **720**. At this time, the light which is extracted to the outside of the light emitting diode **720** is reflected from the first reflective electrode RE**1** and the second reflective electrode RE**2** below the light emitting diode **720** and the reflective partition **830** on the side portion of the light emitting diode **720** to travel to the upper portion of the light emitting diode **720**. The first reflective electrode RE**1** or the second reflective electrode RE**2** extends while overlapping a part of the reflective partition **830** at the left and right of the light emitting diode **720** to overlap the entire area of the light emitting diode **720** to reflect light emitted from the light emitting diode **720** and light reflected from the reflective partition **830**. For example, the first reflective electrode RE**1** can be disposed so as to overlap the entire light emitting diode **720** and the entire reflective partition **830**. Further, at least some of light emitted from the light emitting diode **720** or light reflected from the reflective partition **830** can be reflected to the upper portion of the substrate **110** by the first reflective electrode RE**1**. Accordingly, the light extraction efficiency of the light emitting diode **720** can be improved and the luminance of the display device **800** can be improved.

[0179] FIG. **9** is a cross-sectional view of a display device according to still another example embodiment of the present disclosure. The difference between a display device **900** of FIG. **9** and the display device **800** of FIG. **8** is a reflective partition **930**, but the other configuration is substantially the same, so that a redundant description will be omitted or may be briefly provided.

[0180] Referring to FIG. **9**, the reflective partition **930** is disposed on the second planarization layer **118**. The reflective partition **930** can be disposed on the second planarization layer **118** to enclose the light emitting diode **720**. The reflective partition **930** can reflect some light which is directed to the side surface of the light emitting diode **720** to the upper portion of the light emitting diode **720**.

[0181] A partition layer **931** of the reflective partition **930** is disposed on the second planarization layer **118**. For example, the partition layer **931** can be formed together when the second planarization layer **118** is formed and the partition layer **931** is integrally formed with the second planarization layer **118**. The partition layer **931** is disposed to protrude from the top surface of the second planarization layer **118**.

[0182] Further, after forming the second planarization layer **118** and the partition layer **931**, the reflective layer **932** is formed on the partition layer **931** and the third planarization layer **119** can be formed on the reflective layer **932**. The reflective layer **932** can be disposed so as to cover a top surface and a side surface of the partition layer **931**.
[0183] The third planarization layer **119** is disposed on the reflective partition **930**. The third planarization layer **119** can be disposed so as to cover upper portions of the reflective partition **930** and the light emitting diode **720**.
[0184] Accordingly, in the display device **900** according to still another example embodiment of the present disclosure, the second planarization layer **118** and the partition layer **931** of the reflective partition **930** are formed together to simplify the manufacturing process of the reflective partition **930**. For example, the second planarization layer **118** and the partition layer **931** can be formed by the same process and the same material. Therefore, the manufacturing process of the partition layer **931** of the reflective partition **930** is integrated with the manufacturing process of the second planarization layer **118** to simplify the manufacturing process of the reflective partition **930**.
[0185] The example embodiments of the present disclosure can also be described as follows:
[0186] According to an aspect of the present disclosure, a display device includes a substrate in which a plurality of sub pixels is defined, and a light emitting diode which is disposed in each of the plurality of sub pixels and has a first uneven structure, and the light emitting diode includes a first semiconductor layer which has a first uneven structure formed on a bottom surface, an emission layer on the first semiconductor layer, a second semiconductor layer on the emission layer, a first electrode disposed on at least a part of a side surface of the first semiconductor layer, and a second electrode disposed on the second semiconductor layer.
[0187] A bottom surface of the first semiconductor layer can have a semi-polarity or a non-polarity.
[0188] The first electrode can include a first electrode layer which is in contact with a side surface of the first semiconductor layer, and a second electrode layer disposed on the first electrode layer, and the second electrode layer can be formed of a ferromagnetic material.
[0189] The first semiconductor layer can partially protrude from the emission layer and the second semiconductor layer and one end portion of the first electrode layer and one end portion of the second electrode layer can be disposed on a top surface of a protruding part of the first semiconductor layer.
[0190] The other end portion of the first electrode layer and the other end portion of the second electrode layer can be disposed to be spaced apart from a lower edge of the first semiconductor layer.
[0191] The light emitting diode can further include a second uneven structure formed in a remaining part of the side surface of the first semiconductor layer excluding a part in which the first electrode is disposed.
[0192] The light emitting diode can further include an encapsulation film which encloses the second semiconductor layer, the emission layer, and at least a part of the first semiconductor layer and the encapsulation film can cover the remaining part of the side surface of the first semiconductor layer excluding a part in which the second uneven structure and the first electrode are disposed.
[0193] An end portion of the encapsulation film can be disposed to be spaced apart from a lower edge of the first semiconductor layer.
[0194] The first uneven structure and the second uneven structure can be exposed from the first electrode and the encapsulation film.
[0195] The first uneven structure and the second uneven structure can be irregular uneven structures.
[0196] The display device can further include a reflective partition disposed on each of the plurality of sub pixels so as to enclose the light emitting diode, and the reflective partition can include a partition layer which encloses the light emitting diode, and a reflective layer disposed on a top

surface and a side surface of the partition layer.
[0197] The reflective partition can be configured to reflect light extracted to a side surface of the first semiconductor layer to an upper portion of the light emitting diode.
[0198] The display device can further include a first planarization layer disposed between the substrate and the light emitting diode, an adhesive layer disposed between the first planarization layer and the light emitting diode, a first connection electrode which is disposed on the adhesive layer and is electrically connected to the first electrode of the light emitting diode, a second planarization layer disposed on the first connection electrode and the light emitting diode, a third planarization layer disposed on the second planarization layer, and a second connection electrode which is disposed on the third planarization layer and is electrically connected to the second electrode of the light emitting diode. The reflective layer of the reflective partition can be disposed to be spaced apart from the second connection electrode with the third planarization layer therebetween.
[0199] The partition layer of the reflective partition can be disposed on the adhesive layer and the first connection electrode.
[0200] The partition layer of the reflective partition can be disposed to protrude from a top surface of the second planarization layer.
[0201] According to an aspect of the present disclosure, a method of manufacturing a display device includes placing a wafer and a plurality of light emitting diodes formed on one surface of the wafer in a chamber filled with a solution, separating the plurality of light emitting diodes from the wafer, placing an assembling substrate above the chamber, and moving the plurality of light emitting diodes to the assembling substrate by placing a magnet on n the assembling substrate and self-assembling the plurality of light emitting diodes on the assembling substrate.
[0202] The solution in the chamber can include at least one of de-ionized water and an alkaline etchant.
[0203] An uneven structure can be formed on bottom surfaces of the plurality of light emitting diodes by the solution in the chamber.
[0204] Although the example embodiments of the present disclosure have been described in detail with reference to the accompanying drawings, the present disclosure is not limited thereto and can be embodied in many different forms without departing from the technical concept of the present disclosure. Therefore, the example embodiments of the present disclosure are provided for illustrative purposes only but not intended to limit the technical concept of the present disclosure. The scope of the technical concept of the present disclosure is not limited thereto. Therefore, it should be understood that the above-described example embodiments are illustrative in all aspects and do not limit the present disclosure. All the technical concepts in the equivalent scope of the present disclosure should be construed as falling within the scope of the present disclosure.

Claims

1. A display device, comprising: a substrate in which a plurality of sub pixels is defined; and a light emitting diode disposed in each of the plurality of sub pixels and having a first uneven structure, wherein the light emitting diode includes: a first semiconductor layer having the first uneven structure formed on a bottom surface of the first semiconductor layer; an emission layer on the first semiconductor layer; a second semiconductor layer on the emission layer; a first electrode disposed on at least a part of a side surface of the first semiconductor layer; and a second electrode disposed on the second semiconductor layer.
2. The display device according to claim 1, wherein the bottom surface of the first semiconductor layer has a semi-polarity or a non-polarity.
3. The display device according to claim 1, wherein the first electrode includes: a first electrode layer in contact with a side surface of the first semiconductor layer; and a second electrode layer

disposed on the first electrode layer, and wherein the second electrode layer includes a ferromagnetic material.
4. The display device according to claim 3, wherein the first semiconductor layer partially protrudes from the emission layer and the second semiconductor layer, and one end portion of the first electrode layer and one end portion of the second electrode layer are disposed on a top surface of a protruding part of the first semiconductor layer.
5. The display device according to claim 4, wherein another end portion of the first electrode layer and another end portion of the second electrode layer are disposed to be spaced apart from a lower edge of the first semiconductor layer.
6. The display device according to claim 3, wherein the light emitting diode further includes a second uneven structure formed in a remaining part of the side surface of the first semiconductor layer excluding a part in which the first electrode is disposed.
7. The display device according to claim 6, wherein the light emitting diode further includes an encapsulation film enclosing the second semiconductor layer, the emission layer, and at least a part of the first semiconductor layer, and wherein the encapsulation film covers the remaining part of the side surface of the first semiconductor layer excluding a part in which the second uneven structure and the first electrode are disposed.
8. The display device according to claim 7, wherein an end portion of the encapsulation film is disposed to be spaced apart from a lower edge of the first semiconductor layer.
9. The display device according to claim 7, wherein the first uneven structure and the second uneven structure are exposed from the first electrode and the encapsulation film.
10. The display device according to claim 7, wherein the first uneven structure and the second uneven structure are irregular uneven structures.
11. The display device according to claim 6, further comprising: a reflective partition disposed on each of the plurality of sub pixels so as to enclose the light emitting diode, wherein the reflective partition includes: a partition layer enclosing the light emitting diode; and a reflective layer disposed on a top surface and a side surface of the partition layer.
12. The display device according to claim 11, wherein the reflective partition is configured to reflect light extracted to a side surface of the first semiconductor layer to an upper portion of the light emitting diode.
13. The display device according to claim 11, further comprising: a first planarization layer disposed between the substrate and the light emitting diode; an adhesive layer disposed between the first planarization layer and the light emitting diode; a first connection electrode disposed on the adhesive layer and electrically connected to the first electrode of the light emitting diode; a second planarization layer disposed on the first connection electrode and the light emitting diode; a third planarization layer disposed on the second planarization layer; and a second d connection electrode disposed on the third planarization layer and electrically connected to the second electrode of the light emitting diode, wherein the reflective layer of the reflective partition is disposed to be spaced apart from the second connection electrode with the third planarization layer therebetween.
14. The display device according to claim 13, wherein the partition layer of the reflective partition is disposed on the adhesive layer and the first connection electrode.
15. The display device according to claim 13, wherein the partition layer of the reflective partition is disposed to protrude from a top surface of the second planarization layer.
16. A method of manufacturing a display device, the method comprising: placing a wafer and a plurality of light emitting diodes formed on one surface of the wafer in a chamber filled with a solution; separating the plurality of light emitting diodes from the wafer; placing an assembling substrate on the chamber; and moving the plurality of light emitting diodes to the assembling substrate by placing a magnet on the assembling substrate and self-assembling the plurality of light emitting diodes on the assembling substrate.
17. The method of manufacturing the display device according to claim 16, wherein the solution in

the chamber includes at least one of de-ionized water and an alkaline etchant.

18. The method of manufacturing the display device according to claim 17, wherein an uneven structure is formed on bottom surfaces of the plurality of light emitting diodes by the solution in the chamber.

19. A display device, comprising: a substrate; and a plurality of sub pixels disposed on the substrate; wherein each of the plurality of sub pixels comprises a light emitting diode including: a first semiconductor layer having a bottom surface, a top surface and a side surface; an emission layer on the top surface; and a first electrode disposed on at least a part of the side surface, and wherein the bottom surface of the first semiconductor layer has a first uneven structure.

20. The display device according to claim 19, wherein the first uneven structure is an uneven structure formed by wet-etching using an alkaline etchant.